# A1.2 Nucleic acids

| NAME | DATE | TOTAL |
|---|---|---|
| | | _____ / 36 |

**About 35 minutes,** or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

## Section A: Recall

8 MARKS

1 STATE the molecule that carries genetic information in every living organism. [A1.2.1] [1]

2 LIST the components of a nucleotide. [A1.2.2] [1]

3 NAME the base found in RNA but not in DNA. [A1.2.4] [1]

4 STATE the type of reaction that links RNA nucleotides. [A1.2.5] [1]

5 STATE the type of bond between complementary bases. [A1.2.6] [1]

6 STATE three differences between DNA and RNA. [A1.2.7] [3]

## Section B: Understanding and drawing

16 MARKS

7 DRAW a section of DNA with four base pairs, showing the bases, the backbone and the bonds. [A1.2.6] [5]

8 **SKETCH** deoxyribose and ribose, to show the difference between them. [A1.2.7] [2]

9 **EXPLAIN** what is meant by the sugar-phosphate backbone, and why it is strong. [A1.2.3] [3]

10 **EXPLAIN** how complementary base pairing allows genetic information to be replicated. [A1.2.8] [3]

11 **EXPLAIN** why almost identical genetic codes in all organisms is evidence of common ancestry. [A1.2.10] [3]

## Section C: Working with data

6 MARKS

*All data in this section are invented for practice. The patterns are realistic.*

12 [A1.2.6, A1.2.9]

A section of double-stranded DNA had 1000 base pairs. It was found to contain 280 adenine bases on strand 1 and 220 adenine bases on strand 2.

a **CALCULATE** the percentage of all bases in this section that are adenine. [2]

b **DEDUCE** the percentage of guanine. [2]

c **CALCULATE** the number of possible sequences of a strand 6 bases long. [2]

## Section D: Longer answer

6 MARKS

13 **OUTLINE** how the structure of DNA allows it to store a very large quantity of information accurately. [6]

[A1.2.3-9]